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Studierichting: Informatica

Oefeningenreeks 2B nr. 1.8

6.7) $4 \operatorname{Bgtan} \frac{1}{5} - \operatorname{Bgtan} \frac{1}{239}$ toon aan dat de uitdrukking gelijk is aan $\frac{\pi}{4}$.

$$\begin{aligned} & 4 \operatorname{Bgtan} \frac{1}{5} - \operatorname{Bgtan} \frac{1}{239} \\ &= 2 \operatorname{Bgtan} \left(\frac{\frac{1}{5} + \frac{1}{5}}{1 - \frac{1}{5} \cdot \frac{1}{5}} \right) - \operatorname{Bgtan} \left(\frac{1}{239} \right) \\ &= 2 \operatorname{Bgtan} \left(\frac{10}{24} \right) - \operatorname{Bgtan} \left(\frac{1}{239} \right) \\ &= \operatorname{Bgtan} \left(\frac{\frac{10}{24} + \frac{10}{24}}{1 - \frac{10}{24} \cdot \frac{10}{24}} \right) - \operatorname{Bgtan} \left(\frac{1}{239} \right) \\ &= \operatorname{Bgtan} \left(\frac{120}{119} \right) - \operatorname{Bgtan} \left(\frac{1}{239} \right) \\ &= \operatorname{Bgtan} \left(\frac{\frac{120}{119} - \frac{1}{239}}{1 + \frac{120}{119} \cdot \frac{1}{239}} \right) \\ &= \operatorname{Bgtan} \left(\frac{\frac{28680}{28441} - \frac{119}{28441}}{\frac{28441}{28441} + \frac{120}{28441}} \right) \\ &= \operatorname{Bgtan}(1) = \frac{\pi}{4} \end{aligned}$$

References